Road bicycle concept

A coherent 700c layout, transparent assumptions and a reproducible CAD handoff.



SCHEMATIC AXONOMETRIC VIEW

Vector projection from the master coordinates; reference components are illustrative.

DESIGN ENVELOPE

990 mm
Wheelbase

73.0° / 73.5°
Head / seat tube angle

410 mm
Chainstay, axle to BB

70 mm
Bottom bracket drop

Medium-size welded round-tube steel concept. Tube grade, detailed joints and rider fit remain unselected.

Rim-brake layout and 100 / 130 mm hub spacing are assumed. No supplier CAD has been imported.

01 Geometry

Shared coordinates drive the frame layout, nominal tubes and component envelopes.

02 Exchange model

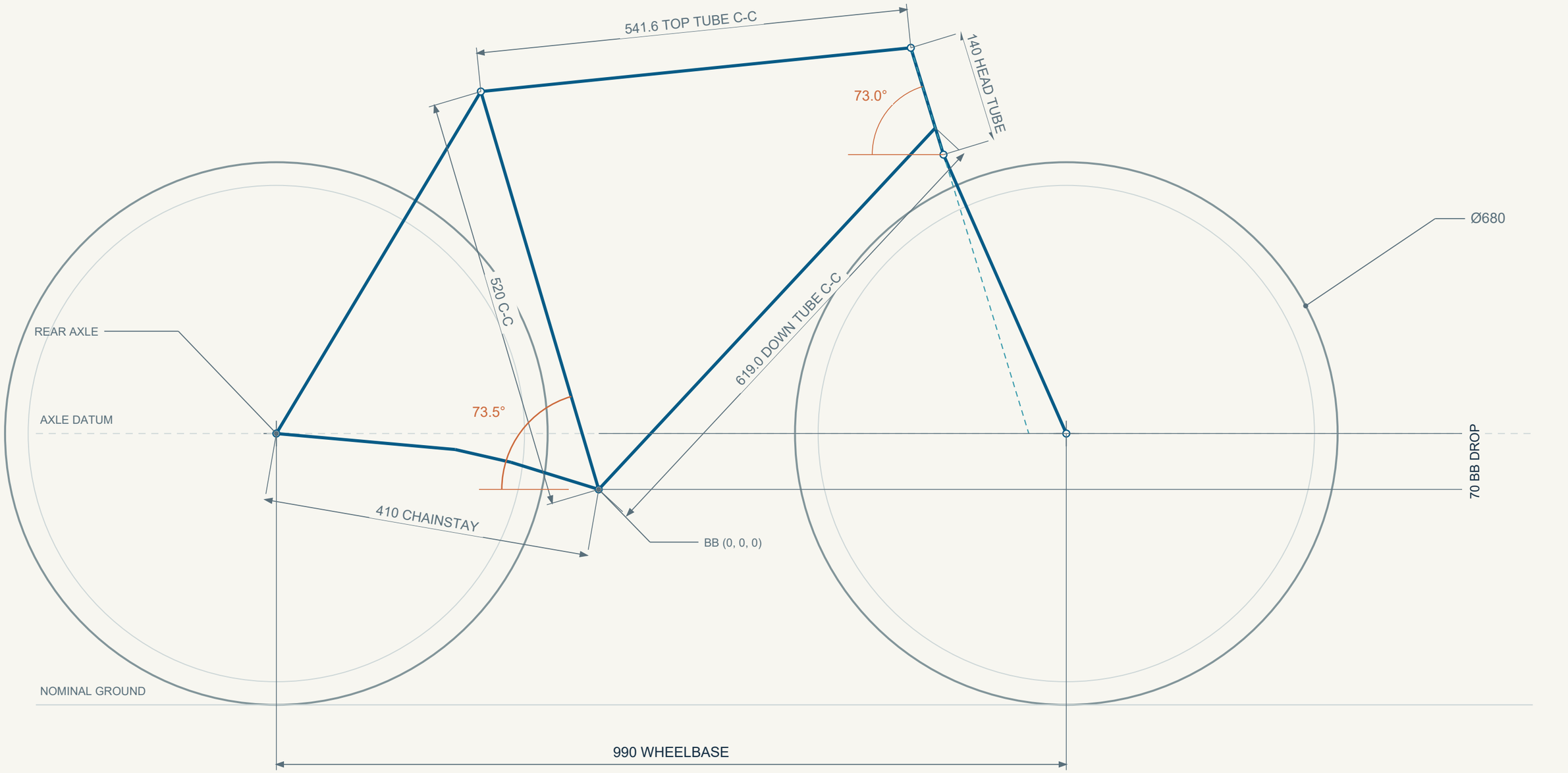
STEP geometry supports review and import into a mechanical CAD environment.

03 Native handoff

SolidWorks creation is scripted but unexecuted. Native features and mates require local verification.

Master geometry / side elevation

Right-side schematic elevation | Dimensions define the master skeleton, not tube cut lengths.

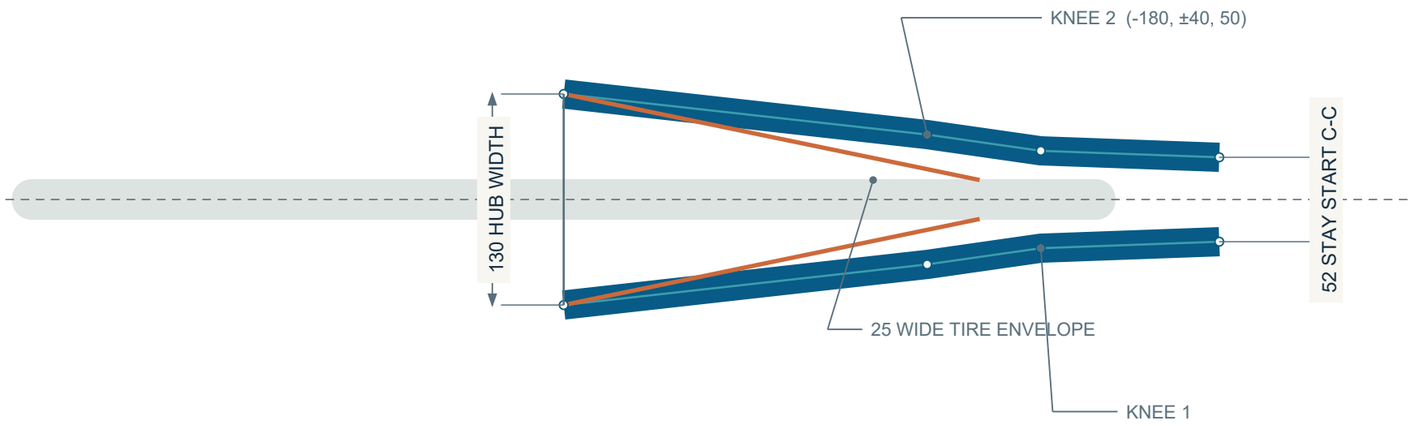


REACH	STACK	EFFECTIVE TOP TUBE	FORK OFFSET	GEOMETRIC TRAIL	BB GROUND HEIGHT
391.2 mm	553.4 mm	555.1 mm	45 mm	56.9 mm	270 mm

Rear triangle / clearance intent

Orthographic projections establish the nominal path. Three-dimensional clearance results are reported separately.

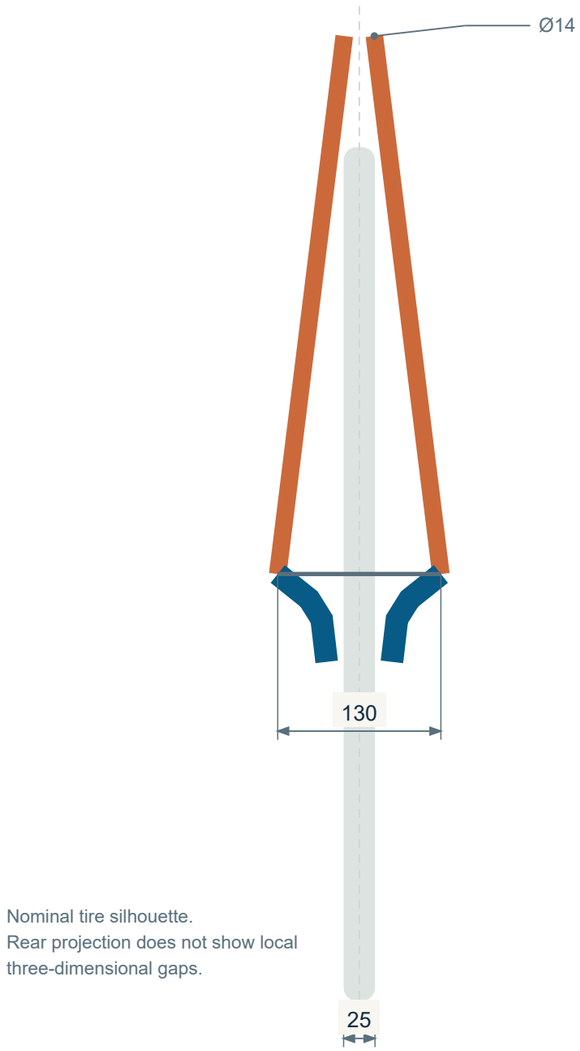
TOP PROJECTION / X-Y



Blue: Ø18 chainstay path Orange: seatstay projection

Plan-view overlap does not indicate collision; Z must be included in the check.

REAR PROJECTION / Y-Z



Nominal tire silhouette.
Rear projection does not show local three-dimensional gaps.

DESIGN TARGET: AT LEAST 6 mm TIRE-TO-FRAME GAP

7.378 mm	13.163 mm	23.644 mm
REAR TIRE / CHAINSTAY	REAR TIRE / SEAT TUBE	FRONT TIRE / DOWN TUBE
OCCT nominal BRep distance	OCCT nominal BRep distance	OCCT nominal BRep distance

OCCT boundary-representation minimum distances use nominal tubes and the specified 680 × 25 mm tire envelope. They exclude manufacturing tolerances, wheel flex, inflation growth, debris and supplier-specific tire shape. The plan projections are not clearance sections.

The brief describes ~680 mm as a true 700x25c envelope. Here it is a project assumption: 25-622 identifies nominal tire width and bead-seat diameter; actual mounted dimensions depend on the tire/rim combination.

Coordinate definition / member schedule

Single source of truth: design_parameters.json, reference_points.csv and tube_schedule.csv.

MASTER POINTS / MM

POINT	X	Y	Z
BB	0.00	0.00	0.00
RearAxle	-403.98	0.00	70.00
FrontAxle	586.02	0.00	70.00
ForkCrown	435.61	0.00	408.05
HeadLower	432.10	0.00	419.53
HeadUpper	391.17	0.00	553.41
SeatTop	-147.69	0.00	498.59
DownTubeHead	421.87	0.00	453.00
ChainStartL	0.00	26.00	0.00
ChainKnee1L	-110.00	30.00	34.00
ChainKnee2L	-180.00	40.00	50.00
RearDropoutL	-403.98	65.00	70.00
SeatStayStartL	-147.69	12.00	498.59
ForkShoulderL	435.61	37.00	408.05
FrontDropoutL	586.02	50.00	70.00
ChainStartR	0.00	-26.00	0.00
ChainKnee1R	-110.00	-30.00	34.00
ChainKnee2R	-180.00	-40.00	50.00
RearDropoutR	-403.98	-65.00	70.00
SeatStayStartR	-147.69	-12.00	498.59
ForkShoulderR	435.61	-37.00	408.05
FrontDropoutR	586.02	-50.00	70.00

PROVENANCE

The brief controls wheelbase range, tube angles, BB drop and chainstay range. All other values are explicit design assumptions. Left / right stay paths are mirrored about Y = 0.

TUBE SCHEDULE / NOMINAL SECTION

MEMBER	OD	WALL	C-LENGTH
SeatTube	28.6	1.0	520.0
TopTube	28.6	0.9	541.6
DownTube	35.0	1.1	619.0
HeadTube	38.0	2.0	140.0
ChainstayL1	18	1	115.2
ChainstayL2	18	1	72.5
ChainstayL3	18	1	226.3
SeatstayL	14	0.9	502.2
ChainstayR1	18	1	115.2
ChainstayR2	18	1	72.5
ChainstayR3	18	1	226.3
SeatstayR	14	0.9	502.2

All lengths are centerline references before coping. They are not fabrication cut lengths. Nominal walls are modelling inputs and have not been strength-qualified.

CONCEPT DATUM STRATEGY

Origin: bottom bracket center. X points forward, Y to the rider's left and Z upward. The frame center plane is Y = 0. Axles are parallel to Y; both nominal axle centers lie at Z = 70.

Proposed inspection references: BB axis, frame median plane, rear axle axis. Detailed datum features, fit tolerances, dropout parallelism and weld-distortion allowances require a manufacturing design.

Assembly plan / component interfaces

SolidWorks implementation intent. Mates below are specified for the native handoff and have not been executed or validated.

ASSEMBLY CONSTRAINT PLAN

COMPONENT	MATE INTENT	REMAINING DOF	IMPLEMENTATION NOTE
Frame	Fix origin / orientation	0	Use the skeleton as the grounded reference
Fork + steering	Concentric + axial position	1 rotation	Do not lock the steering concentric mate
Front wheel	Concentric + width/center	1 rotation	Center in nominal 100 mm front dropout gap
Rear wheel	Concentric + width/center	1 rotation	Center in nominal 130 mm rear dropout gap
Crank axle	Concentric + axial position	1 rotation	Use BB shell axis, nominal 68 mm width
Crank / rear drive	Illustrative gear relation	Coupled	50:17 example; verify direction and ratio
Cockpit / saddle	Placement / rigid group	0 intended	Reference geometry only; final interfaces TBD

A gear mate is a simplified kinematic relation. It does not reproduce chain contact, derailleur shifting, backlash or freehub coasting. Wheel rotation, steering and drivetrain motion must be exercised in SolidWorks after native mate creation.

PURCHASED-COMPONENT REGISTER / UNSELECTED

SYSTEM	CURRENT ASSUMPTION	REQUIRED BEFORE DETAIL DESIGN
Tire + rim	25-622 class; 680 × 25 envelope	Verify mounted width/OD and rim compatibility
Front / rear hubs	100 / 130 mm nominal spacing	Select axle format, dropouts and hub CAD
Headset + fork	45 offset; 370 axle-to-crown	Confirm steerer, bearing seats and headset stack
Bottom bracket + crank	68 shell; 172.5 crank; 50T example	Select shell standard, chainline and pedal clearance
Brake / drivetrain / cockpit	Rim-brake concept; 17T rear example	Add mounts, cable routes and supplier interfaces

No commercial parts are claimed to be sourced, licensed, purchase-ready or supplier-approved. Supplier models must be checked for revision, units, origin, envelope accuracy and permitted use before integration.

Validation record / release boundary

Nominal model verification and exchange-file evidence. Native CAD completion and manufacturing release remain open.

EVIDENCE AND STATUS

CHECK	RESULT	TARGET	STATUS / LIMITATION
Rear tire to chainstays	7.378 mm	>= 6 mm	Pass at nominal geometry
Rear tire to seatstays	10.572 mm	>= 6 mm	Pass at nominal geometry
Rear tire to seat tube	13.163 mm	>= 6 mm	Pass at nominal geometry
Front tire to down tube	23.644 mm	>= 6 mm	Pass at nominal geometry
Front tire to fork	13.470 mm	>= 6 mm	Pass at nominal geometry
Chainring envelope to chainstays	4.344 mm	>= 3 mm	Pass at nominal geometry; internal 3 mm target
Chain strands to chainstays	28.447 mm	>= 1 mm	Pass at nominal geometry; two straight strands only
Full crank-arm sweep to frame	17.485 mm	>= 3 mm	Pass at nominal geometry; conservative crank-arm disks
Front tire steering samples to frame	23.644 mm	>= 6 mm	Pass at discrete steering samples
STEP export / reimport	9 components	Valid BReps	Passed; assembly valid
SolidWorks rebuild / mates	Not executed	Native review	No native feature or mate validation available
Fit / structural performance	Not evaluated	Engineering	Rider fit, material, joints, loads and physical tests required
Manufacturing drawing release	Not released	Detail design	Tolerances, welds, interfaces and inspection plan pending

Distances are OCCT BRep minimum distances. Steering uses -35° to +35° at 5° increments; no continuous sweep is proven. Chainring / crank / chain-strand targets of 3 / 3 / 1 mm are internal concept criteria.

CAD COMPLETION SEQUENCE

1 Import and rebuild

Open the STEP model or execute the supplied native-creation automation. Confirm mm units, origins, orientation and reference planes.

2 Resolve interfaces

Select actual tires, hubs, drivetrain, headset and brakes. Replace reference envelopes and complete joints, bends, dropouts and bearing details.

3 Validate and release

Resolve intentional overlaps, run static and dynamic interference, establish tolerance budgets and structural tests, then create controlled manufacturing drawings.

REFERENCES / DESIGN INTERPRETATION

Project brief: original_brief.md (provided by the user). Geometry and modelling assumptions: design_parameters.json. Nominal tire sizing guidance: Schwalbe Technology FAQ and Tire Sizes Overview. CAD exchange workflow: CadQuery import/export documentation.

Schwalbe sizing FAQ

Schwalbe tire-size overview

CadQuery import/export